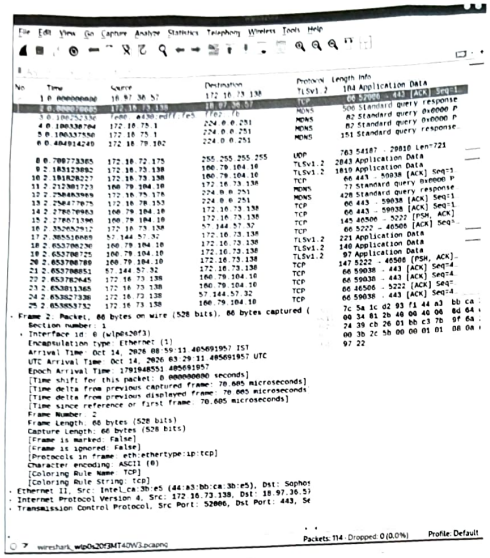
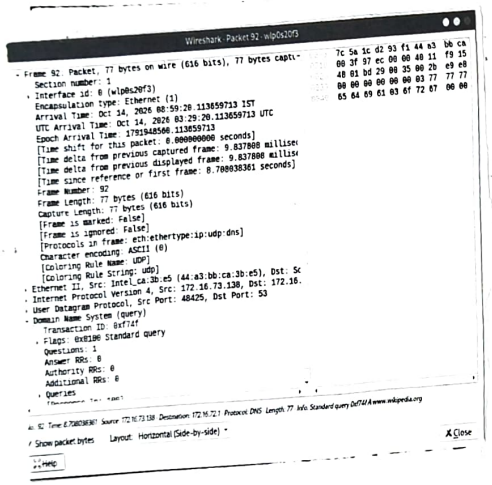
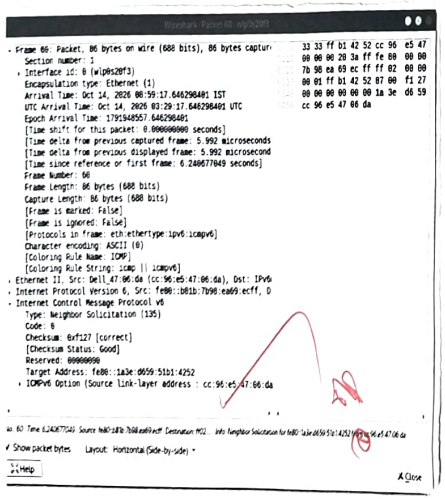






Exp no: 14

Date: 26/9

PACKET SNIFFING USING RAW SOCKETS

Aim: To capture packets and display packet header information using raw sockets.

Program:

raw_sniffer.py

```
import socket
import struct
import sys

def main():
    try:
        sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
    except OSError as exc:
        print("Could not create raw socket:", exc)
        print("On Linux, try running with sudo on an authorized test machine.")
        return
    print("Capturing IP packets. Press Ctrl+C to stop.")
    try:
        while True:
            packet, address = sniffer.recvfrom(65535)
```

```
        if len(packet) < 20:
            continue

        version_ihl = packet[0]
        version = version_ihl >> 4
        ihl = (version_ihl & 0x0F) * 4
        total_length = struct.unpack("!H", packet[2:4])[0]
        protocol = packet[9]
        source = socket.inet_ntoa(packet[12:16])
        destination = socket.inet_ntoa(packet[16:20])

        if version == 4 and ihl >= 20:
            print("-" * 50)
            print("Source IP :", source)
            print("Destination IP :", destination)
            print("Protocol number :", protocol)
            print("IP Header Size :", ihl, "bytes")
            print("Total Length :", total_length, "bytes")
    except KeyboardInterrupt:
        print("\n Capture Stopped.")
    finally:
        sniffer.close()

if __name__ == "__main__":
    main()
```

Result:

Packet header information was captured and displayed in an authorized test environment.